

Considering the MIPS64 architecture presented in the following:

- [illegible]

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Question 2

Considerando il programma precedente e l'architettura del processore superscalare descritto in seguito; completare la tabella relativa alle prime 2 iterazioni.

Processor architecture:

- Issue 2 instructions per clock cycle
- jump instructions require 1 issue
- handle 2 instructions commit per clock cycle
- timing facts for the following separate functional units:
 - i. 1 Memory address 1 clock cycle
 - ii. 1 Integer ALU 1 clock cycle
 - iii. 1 Jump unit 1 clock cycle
 - iv. 1 FP multiplier unit, which is pipelined: 6 stages
 - v. 1 FP divider unit, which is not pipelined: 7 clock cycles
 - vi. 1 FP Arithmetic unit, which is pipelined: 2 stages
- Branch prediction is always correct
- There are no cache misses
- There are 2 CDB (Common Data Bus).

# iteration		Issue	EXE	MEM	CDB x2	COMMIT x2
1	l.d f1,v1(r1)	1	2m	3	4	5
1	l.d f2,v2(r1)	1	3m	4	5	6
1	l.d f3,v3(r1)	2	4m	5	6	7
1	l.d f4,v4(r1)	2	5m	6	7	8
1	div.d f5,f1,f2	3	6d		13	14
1	s.d f5,v5(r1)	3	6m			14
1	mul.d f6,f1,f2	4	6x		12	15
1	div.d f7,f3,f4	4	13d		20	21
1	add.d f1,f6,f7	5	21a		23	24
1	s.d f1,v6(r1)	5	7m			24
1	daddui r1,r1,8	6	7i		8	25
1	daddi r2,r2,-1	6	8i		9	25
1	bnez r2,loop	7	10j			26
2	l.d f1,v1(r1)	8	9m	10	11	26
2	l.d f2,v2(r1)	8	10m	11	12	27
2	l.d f3,v3(r1)	9	11m	12	13	27
2	l.d f4,v4(r1)	9	12m	13	14	28
2	div.d f5,f1,f2	10	20d		27	28
2	s.d f5,v5(r1)	10	13m			29
2	mul.d f6,f1,f2	11	13x		19	29
2	div.d f7,f3,f4	11	27d		34	35
2	add.d f1,f6,f7	12	35a		37	38
2	s.d f1,v6(r1)	12	14m			38
2	daddui r1,r1,8	13	14i		15	39
2	daddi r2,r2,-1	13	15i		16	39
2	bnez r2,loop	14	17j			40